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Inventor(s)	Vargas Garcia; Carmen Frida et al.

Air register assembly

Abstract

An air register assembly for a vehicle includes a barrel and vanes operably coupled to the barrel. A knob assembly is operably coupled to the barrel and the vanes. The knob assembly includes a support body, at least one absorbent basket selectively coupled to the support body, and a cover selectively coupled over the at least one absorbent basket. The cover is configured to rotate between an opened position and a closed position. The opened position is configured to provide fluid communication with the at least one absorbent basket via apertures defined by the cover.

Inventors: Vargas Garcia; Carmen Frida (Cuautitlan Izcalli, MX), Tena Han; Luis Adrian (Tlalnepantla de Baz, MX)

Applicant: Ford Global Technologies, LLC (Dearborn, MI)

Family ID: 1000008822307

Assignee: Ford Global Technologies, LLC (Dearborn, MI)

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Primary Examiner: Bosques; Edelmira

Assistant Examiner: Mallon; Brett Peterson

Attorney, Agent or Firm: Price Heneveld LLP

Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to an air register assembly. More specifically, the present disclosure relates to an air register assembly for a vehicle.

BACKGROUND OF THE DISCLOSURE

(2) Many vehicles include a heating, ventilation, and air conditioning system to direct air into a passenger cabin. Air vents may direct air from the heating, ventilation, and air conditioning system to various locations within the passenger cabin.

SUMMARY OF THE DISCLOSURE

(3) According to at least one aspect of the present disclosure, a vehicle air register assembly includes a housing defining an outlet. A barrel is disposed within the housing proximate to the outlet. The barrel is rotatably coupled to the housing. A frame is coupled to the barrel. Vanes are rotatably coupled to the frame proximate to the outlet. A knob assembly is coupled to the frame adjacent to the vanes. The knob assembly includes a support body defining an interior. Pivots extend from the support body to engage the frame. An engagement shaft extends through the interior of the support body. An absorbent basket is coupled to the engagement shaft. The absorbent basket is configured to absorb a fluid for providing a scent. A cover is selectively coupled over the absorbent basket. The cover defines apertures to provide fluid communication with the absorbent basket.

(4) According to another aspect of the present disclosure, an air register assembly for a vehicle includes a barrel and vanes operably coupled to the barrel. A knob assembly is operably coupled to the barrel and the vanes. The knob assembly includes a support body, at least one absorbent basket selectively coupled to the support body, and a cover selectively coupled over the at least one absorbent basket. The cover is configured to rotate between an opened position and a closed position. The opened position is configured to provide fluid communication with the at least one absorbent basket via apertures defined by the cover.

(5) According to another aspect of the present disclosure, a knob assembly for a vehicle air register includes a support body defining an interior. An engagement shaft extends through the interior. An absorbent basket includes a base and an outer shell defining an interior. The absorbent basket is coupled to the engagement shaft. An absorbing material is disposed within the interior of the absorbent basket. A cover is selectively coupled over the absorbent basket. The cover defines apertures and is operable between an opened position and a closed position.

(6) These and other aspects, objects, and features of the present disclosure will be understood and appreciated by those skilled in the art upon studying the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following is a description of the figures in the accompanying drawings. The figures are not necessarily to scale, and certain features and certain views of the figures may be shown exaggerated in scale or in schematic in the interest of clarity and conciseness.

(2) In the drawings:

(3) FIG. 1 is a partial front perspective view of a dashboard with air register assemblies, according to the present disclosure;

(4) FIG. 2 is a side perspective view of an air register assembly coupled to ducting, according to the present disclosure;

(5) FIG. 3 is a side perspective view of an air register assembly, according to the present disclosure;

(6) FIG. 4 is a cross-sectional view of the air register assembly of FIG. 3, taken along line IV-IV, according to the present disclosure;

(7) FIG. 5 is a partially exploded side perspective view of an air register assembly with a knob assembly including an absorbent basket and a cover, according to the present disclosure;

(8) FIG. 6 is a partially exploded side perspective view of a knob assembly including a cover

disengaged from an absorbent basket, according to the present disclosure;

(9) FIG. 7 is a side perspective view of a knob assembly for an air register assembly, according to the present disclosure;

(10) FIG. 8 is a partially exploded side perspective view of an air register assembly with a knob assembly having a removable cover and interchangeable absorbent baskets, according to the present disclosure;

(11) FIG. 9 is a side perspective view of an air register assembly with a cover of a knob assembly in an opened position, according to the present disclosure; and

(12) FIG. 10 is a side perspective view of an air register assembly with a cover of a knob assembly in a closed position, according to the present disclosure.

DETAILED DESCRIPTION

(13) Additional features and advantages of the presently disclosed device will be set forth in the detailed description which follows and will be apparent to those skilled in the art from the description, or recognized by practicing the device as described in the following description, together with the claims and appended drawings.

(14) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the concepts as oriented in FIG. 1.

However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(15) As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

(16) The terms “substantial,” “substantially,” and variations thereof as used herein are intended to note that a described feature is equal or approximately equal to a value or description. For example, a “substantially planar” surface is intended to denote a surface that is planar or approximately planar. Moreover, “substantially” is intended to denote that two values are equal or approximately equal. In some embodiments, “substantially” may denote values within about 10% of each other, such as within about 5% of each other, or within about 2% of each other.

(17) As used herein the terms “the,” “a,” or “an,” mean “at least one,” and should not be limited to “only one” unless explicitly indicated to the contrary. Thus, for example, reference to “a component” includes embodiments having two or more such components unless the context clearly indicates otherwise.

(18) In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

(19) With reference to FIGS. 1-10, reference numeral 10 generally designates an air register assembly 10 for a vehicle 12 that includes a housing 14 defining an outlet 16. A barrel 18 is disposed within the housing 14 proximate to the outlet 16. The barrel 18 is rotatably coupled to the

housing **14**. A frame **20** is coupled to the barrel **18**. Rotating vanes **22** are rotatably coupled to the frame **20**. A knob assembly **24** is coupled to the frame **20** adjacent to the rotating vanes **22**. The knob assembly **24** includes a support body **26** that defines an interior **28**. Pivots **30** extend from the support body **26** to engage the frame **20**. The knob assembly **24** also includes an engagement shaft **32** extending through the interior **28** of the support body **26** and an absorbent basket **34** coupled to the engagement shaft **32**. The absorbent basket **34** is configured to absorb fluid for providing a scent. The knob assembly **24** also includes a cover **36** selectively coupled over the absorbent basket **34**. The cover **36** defines apertures **38** to provide fluid communication with the absorbent basket **34**.

(20) The vehicle **12** includes an interior compartment **50** with multiple air register assemblies **10**. For example, as illustrated in FIG. **1**, an instrument panel or dashboard **52** extends within the interior compartment **50** and includes openings **54** for positioning the air register assemblies **10**. The air register assemblies **10** direct air from a heating, ventilation, and air-conditioning (HVAC) system and into the interior compartment **50**. The HVAC system generally cleans, cools, heats, ventilates, and/or dehumidifies air directed into the interior compartment **50**. The air register assemblies **10** are configured to guide, direct, or deflect the air to various locations within the interior compartment **50**.

(21) The vehicle **12** may include multiple air register assemblies **10** disposed in various locations within the interior compartment **50**. The vehicle **12** may be a sedan, a sport-utility vehicle **12**, a van, a truck, a crossover, other styles of wheeled motor vehicles **12**, or other types of vehicles **12**. The vehicle **12** may be a manually operated vehicle **12** (e.g., with a human driver), a fully autonomous vehicle **12** (e.g., with no human driver), or a partially autonomous vehicle **12** (e.g., operated with or without a human driver). Additionally, the vehicle **12** may be utilized for personal and/or commercial purposes, such as ride-providing services (e.g., chauffeuring), ridesharing services, and delivery services. The specific location and number of the air register assemblies **10** within the vehicle **12** may depend on the configuration and uses of the vehicle **12**.

(22) Referring to FIGS. **2** and **3**, the air register assembly **10** includes the housing **14** configured to engage the dashboard **52** to align and retain the air register assembly **10** within the opening in the dashboard **52**. The outlet **16** of the housing **14** is generally coplanar with the dashboard **52**. The housing **14** is coupled to ducting **60** that extends from the HVAC system. An inlet **62** of the housing **14** is disposed adjacent to the ducting **60**. Air is directed from the HVAC system, through the ducting **60**, through the housing **14**, and into the interior compartment **50** (FIG. **1**).

(23) The air register assembly **10** includes the barrel **18** disposed at least partially within the housing **14** proximate the outlet **16**. The barrel **18** is rotatably coupled to the housing **14**, allowing the barrel **18** to rotate about an axis **64** to direct air in different directions. For example, in the illustrated configuration, the barrel **18** is configured to rotate about the vertical axis **64** to direct air in various cross-car directions. The barrel **18** has an elongated shape with a greater longitudinal extent than the lateral extent. In the illustrated configuration, the barrel **18** has a substantially reverse “D”-shape where one lateral side is generally linear and the opposing lateral side is rounded, having a center portion extending between upper and lower angled portions. It is contemplated that the air register assembly **10** may also be rotated, such as about 90°, such that the barrel **18** has a greater lateral extent than longitudinal extent without departing from the teachings herein. The barrel **18** may define any practicable configuration for engaging the housing **14**.

(24) The outlet **16** of the housing **14** generally corresponds with the shape and size of the barrel **18**. In this way, when the barrel **18** is in a center position, the barrel **18** extends along a perimeter of the outlet **16**. When the barrel **18** is in angled positions, the barrel **18** may extend at least partially through the outlet **16**. The passenger within the vehicle **12** may manually adjust the angle of the barrel **18**, which determines a direction of the airflow. The passenger may apply a force to the barrel **18**, the rotating vanes **22**, or the knob assembly **24** to adjust the air register assembly **10** to the desired angle. The housing **14** may have any practicable configuration for directing air into the

interior compartment **50** and allowing movement of the barrel **18**.

(25) Referring still to FIGS. **2** and **3**, the air register assembly **10** includes the frame **20** that extends along an inner surface of the barrel **18** to maximize airflow through the air register assembly **10**. The frame **20** includes opposing retainers **70**, **72** and opposing connector portions **74**, **76** of a support vane **78**. The retainers **70**, **72** extend along the longer lateral sides and generally follow the shape of the sides of the barrel **18**. The first retainer **70** generally has a rounded shape that corresponds with the rounded side of the barrel **18**, and the second retainer **72** generally has a linear shape corresponding with the linear side of the barrel **18**. Additionally, at least one of the retainers **70**, **72** may include a protruding portion **80** proximate to the knob assembly **24** to increase support for applied loads on the knob assembly **24** and the rotating vanes **22**. The retainers **70**, **72** each define recesses or receiving holes **82** configured to receive pivot members **84** of the rotating vanes **22** and the pivots **30** of the knob assembly **24**.

(26) The support vane **78** extends along the longitudinal extent of the barrel **18** and includes the two opposing connector portions **74**, **76**. The connector portions **74**, **76** are configured to couple to the upper and lower sides of the barrel **18**, respectively. Additionally, the connector portions **74**, **76** extend between and are coupled to the retainers **70**, **72**. The connector portions **74**, **76** are generally configured to interlock with ends of the retainers **70**, **72** to provide additional support to the air register assembly **10** and retain the components of the frame **20** in the select configuration along the inner surface of the barrel **18**.

(27) Referring still to FIGS. **2** and **3**, the rotating vanes **22** extend between and are rotatably coupled to the retainers **70**, **72** at the receiving holes **82**. The rotating vanes **22** are generally arranged spaced apart at equidistant intervals from the upper side of the barrel **18** to the lower side of the barrel **18**. Accordingly, the rotating vanes **22** generally extend laterally across the air register assembly **10**. In the illustrated configuration, the receiving holes **82** defined in the first retainer **70** are horizontally aligned with the receiving holes **82** defined in the second retainer **72**. Each vane includes the pivot members **84** that engage the receiving holes **82** and are configured to rotate about respective horizontal axes **86** to direct air at different vertical locations in the interior compartment **50** (FIG. **1**).

(28) While the rotating vanes **22** are generally spaced at equidistant intervals, a larger space is defined between two central rotating vanes **22** to accommodate the knob assembly **24** therebetween. The knob assembly **24** includes the pivots **30** that rotatably couple to the retainers **70**, **72** via the receiving holes **82** in a similar manner as the rotating vanes **22**. The pivots **30** may also operate to guide, direct, or deflect flowing through the air register assembly **10** similar to the rotating vanes **22**. It is also contemplated that the knob assembly **24** may be integrally formed with a center rotating vane **22**.

(29) Referring still to FIG. **3**, as well as FIG. **4**, each rotating vane **22** and the knob assembly **24** includes a linking projection **100** that extends from an upstream portion thereof, respectively. Each linking projection **100** engages a link bar **102** that extends adjacent to the second retainer **72**. The link bar **102** operates to couple each of the rotating vanes **22** and the knob assembly **24** to allow simultaneous rotation of the rotating vanes **22** and the knob assembly **24**. The link bar **102** is generally disposed upstream of and adjacent to the second retainer **72**. It is contemplated that the link bar **102** may be disposed proximate to the first retainer **70** without departing from the teachings herein.

(30) The knob assembly **24** is centrally located in the middle of the arrangement of the rotating vanes **22**. The knob assembly **24** provides a grasping location for the passenger within the vehicle **12** to adjust aspects of their register assembly. As the knob assembly **24** is rotated about the respective horizontal axis **86** that extends between the first and second retainers **70**, **72**, the knob assembly **24** causes the link bar **102** to be vertically adjusted, which then adjusts the remaining rotating vanes **22** by the connections between the linking projections **100** and link bar **102**. Accordingly, manipulation of the knob assembly **24** adjusts the link bar **102**, which simultaneously

adjusts each of the rotating vanes **22** coupled to link bar **102**.

(31) The knob assembly **24** is utilized to adjust the position of the barrel **18** about the respective vertical axis **64** and is also utilized to adjust the rotating vanes **22** about the respective horizontal axes **86**. The knob assembly **24** is adjustable in the cross-car direction (e.g., left and right) to rotate the barrel **18** and adjust the airflow in the cross-car direction. The knob assembly **24** is also vertically adjustable between different angles to rotate the rotating vanes **22** and consequently adjust the airflow in the vertical direction. It is contemplated that the orientation of the air register assembly **10** may be adjusted such that movement of the knob in the cross-car direction adjusts the rotating vanes **22** and movement in the vertical direction adjusts the barrel **18** without departing from the teachings herein.

(32) Referring still to FIGS. **3** and **4**, the knob assembly **24** includes the support body **26** defining the interior **28**, which is substantially hollow. The pivots **30** extend in opposing directions from the support body **26** to engage the retainers **70**, **72**. The engagement shaft **32** extends through the interior **28** of the support body **26**. The engagement shaft **32** extends through an upstream end **110** of the support body **26** to engage a damper adjustment assembly **112** and through a downstream end **114** of the support body **26** to engage the absorbent basket **34**.

(33) The engagement shaft **32** includes a bracket **116** proximate the upstream end **110** of the support body **26**. The bracket **116** extends away from the support body **26** and includes a crossbar **118** for operably coupling the support body **26** to the damper adjustment assembly **112**. The damper adjustment assembly **112** includes a support link **120** rotatably coupled to the crossbar **118**. The rotating engagement between the support link **120** and the crossbar **118** allows the support body **26** to be rotated about the horizontal rotational axis **86** to adjust the positions of the rotating vanes **22** without adjusting the position of the damper adjustment assembly **112**.

(34) Rotation of the cover **36** causes rotation of the engagement shaft **32**, including the crossbar **118**, which, consequently, causes the damper adjustment assembly **112** to be adjusted. The damper adjustment assembly **112** includes a drive gear **130**, which is driven by rotation of the engagement shaft **32**. The drive gear **130** is configured to engage follower gears **132**, **134** of the damper adjustment assembly **112** that are each coupled to a damper plate **136**, **138**, respectively. The rotation of the drive gear **130** causes rotation of the follower gears **132**, **134** to open and close the damper plates **136**, **138**, which open and close an airflow path through the air register assembly **10**.

(35) Referring still to FIGS. **3** and **4**, as well as FIG. **5**, the air register assembly **10** includes the support vane **78** extending along the longitudinal extent of the barrel **18** and forming a portion of the frame **20**. The support vane **78** is generally disposed upstream of the rotating vanes **22** within an interior of the housing **14**. The support vane **78** provides additional support to the air register assembly **10**. The support vane **78** includes the connector portions **74**, **76** that form part of the frame **20**. The connector portions **74**, **76** are vertically aligned with the rotating vanes **22**. A vane portion **150**, **152** extends from each of the connector portions **74**, **76** toward the knob assembly **24**, respectively. In this way, the first vane portion **150** is disposed on one side of the knob assembly **24** (e.g., above the knob assembly **24**), and the second vane portion **152** is disposed on the opposing side of the knob assembly **24** (e.g., below the knob assembly **24**).

(36) The vane portions **150**, **152** generally extend vertically, orthogonal or normal to the rotating vanes **22**. The vane portions **150**, **152** extend upstream from the rotating vanes **22** and are configured to guide, direct, or deflect air flowing through the air register assembly **10** in the cross-car direction. The support vane **78** includes a U-shaped center portion **154** between the vane portions **150**, **152**. The center portion **154** couples the vane portions **150**, **152** to one another and provides space for the knob assembly **24**. Generally, the center portion **154** extends toward one of the lateral sides of the barrel **18** and couples to the barrel **18** proximate to the respective retainer **72**.

(37) Referring to FIGS. **6** and **7**, a downstream portion **160** of the engagement shaft **32** includes locating protrusions **162** and coupling protrusions **164** extending therefrom. The locating protrusions **162** are further downstream compared to the coupling protrusions **164**. The locating

protrusions **162** are configured to align the engagement shaft **32** in the correct orientation relative to the absorbent basket **34**. The absorbent basket **34** includes a base **170** that defines a receiving opening **172** for receiving the downstream portion **160** of the engagement shaft **32**. The receiving opening **172** is shaped to receive the downstream portion **160** and the locating protrusions **162**. The locating protrusions **162** may be elastically deformable to provide an interference fit between the engagement shaft **32** and the absorbent basket **34**.

(38) The absorbent basket **34** includes coupling projections **174** extending from the base **170**. The coupling projections **174** define receiving apertures **176**, and the coupling protrusions **164** on the engagement shaft **32** are configured to be disposed in the receiving apertures **176** to snap engage the engagement shaft **32** with the absorbent basket **34**. The support body **26** includes a recessed region **178** providing space where the coupling projections **174** are configured to engage with the coupling protrusions **164** on the engagement shaft **32**. The base **170** of the absorbent basket **34** is configured to abut the downstream end **114** of the support body **26**. In this way, the absorbent basket **34** extends from the support body **26** and is coupled to the support body **26** through the engagement between the engagement shaft **32** and the coupling projections **174**.

(39) The absorbent basket **34** includes an outer shell **190** coupled to or including the base **170** and extending downstream therefrom. The outer shell **190** and the base **170** define a basket interior **192** for receiving an absorbing material **194**. The outer shell **190** includes a rim **196** adjacent to the base **170**, an outer wall **198** extending from the rim **196**, and an end wall **200** coupled to the outer wall **198** and enclosing the basket interior **192**. The rim **196** is smaller in diameter relative to the base **170**, such that the base **170** provides an annular flange extending beyond the rim **196**. The outer wall **198** is annular and has alternating solid portions **202** and perforated portions **204**. The end wall **200** is generally perforated but may also be solid or a combination thereof.

(40) The absorbing material **194** is disposed within the basket interior **192**. Generally, the absorbent basket **34** is constructed of a plastic material that is sufficiently rigid to maintain a shape but may elastically deform. The absorbing material **194** is generally constructed of a fabric or foam material configured to absorb fluids and liquids. Generally, the absorbing material **194** is configured to receive a scented fluid for providing a scent in the air for passengers around the air register assembly **10**.

(41) Referring still to FIGS. **6** and **7**, the cover **36** is configured to be disposed over the absorbent basket **34**. Generally, the cover **36** snap engages over the absorbent basket **34**. The snap engagement or interference fit between the absorbent basket **34** and the cover **36** may be formed through an engagement between an edge **210** of the cover **36** and one or both of the rim **196** and the base **170**. The cover **36** has an annular wall **212**, which defines a plurality of recessed regions **214**. The recessed regions **214** extend from proximate the edge **210** toward a closed end **216** of the cover **36**. The recessed regions **214** alternate between solid regions **218** and regions defining the apertures **38**. The apertures **38** are configured to provide fluid communication with the absorbent basket **34** and the absorbing material **194** disposed therein.

(42) In various aspects, the cover **36** defines a plurality of internal projections **220** and internal grooves **222** that alternate with one another around an interior of the cover **36**. The internal projections **220** align with the recessed regions **214**, while the internal grooves **222** are disposed between adjacent recessed regions **214**. The internal projections **220** may engage the absorbent basket **34** to couple the cover **36** to the absorbent basket **34**.

(43) Additionally or alternatively, the absorbent basket **34** may define ribs **224**, extending from the base **170** toward the end wall **200** and spaced apart along the rim **196**. In such examples, the ribs **224** are configured to be inserted into the inner grooves of the cover **36**. The ribs **224** may provide alignment for the engagement between the absorbent basket **34** and the cover **36** and may also provide an interference fit to couple the cover **36** to the absorbent basket **34**. Additional shapes of the internal grooves **222** or additional space may be defined to allow for rotation of the cover **36** as described herein. In such examples, additional grooves extending perpendicular to the internal

grooves **222** may allow the cover **36** to rotate relative to the absorbent basket **34** having the ribs **224**.

(44) Referring to FIGS. **8-10**, the absorbent basket **34** is configured to snap engage with the engagement shaft **32** to retain the absorbent basket **34** to the support body **26**. The cover **36** is configured to be selectively disengaged from the absorbent basket **34** by the user to add the scented fluid to the absorbent basket **34**. For example, the scented fluid, illustrated as arrow **230**, is configured to be poured into the absorbent basket **34** through at least one of the perforated portions **204** to be absorbed and retained by the absorbing material **194**. The cover **36** may then be re-coupled over the absorbent basket **34**. When the scent emitted by the scented fluid has weakened or is no longer effective, the user may again remove the cover **36** to add more scented fluid to the absorbent basket **34** and re-couple the cover **36** over the absorbent basket **34**.

(45) Additionally or alternatively, the user may interchange the absorbent basket **34** with additional absorbent baskets **34**. In this way, the user may have a first absorbent basket **232** coupled to the knob assembly **24** for providing a first scent. In order to change the scent or replace the second absorbent basket **234** after a period of time, the user can remove the cover **36** and disengage the first absorbent basket **232** from the engagement shaft **32**. The user may then couple a second absorbent basket **234** to the engagement shaft **32** by snap engaging the coupling projections **174** on the second absorbent basket **234** with the coupling protrusions **164** on the engagement shaft **32**. The cover **36** may then be coupled over the second absorbent basket **234**. In this way, the user may replace the absorbent basket **34** due to wear over time or to interchange the scent. The first and second absorbent baskets **232**, **234** may have a substantially similar structure and may be collectively referred to herein as the absorbent basket **34**.

(46) Each absorbent basket **34** may be configured to receive the scented fluid. Additionally or alternatively, the absorbing material **194** may have a predefined scent. In such examples, the user may enhance or alter the predefined scent by adding the scented fluid.

(47) Referring still to FIGS. **8-10**, the air flowing through the air register assembly **10** is configured to carry the scent from the absorbing material **194** in the absorbent basket **34** into the passenger compartment of the vehicle **12** (FIG. **1**). In this way, airflow through the air register assembly **10** is configured to flow through or past the knob assembly **24** to carry the scent retained within the knob assembly **24**. The user can control the dispersion of the scent by adjusting the cover **36** between an opened position **240** and a closed position **242**. When in the opened position **240**, the apertures **38** defined in the annular wall **212** of the cover **36** are aligned with the perforated portions **204** of the outer shell **190** of the absorbent basket **34**. Accordingly, the apertures **38** allow fluid communication with the absorbing material **194** through the apertures **38** and perforated portions **204** of the outer shell **190**. The fluid communication allows the scent to be carried by air flowing through the air register assembly **10**.

(48) As illustrated in FIG. **10**, if the user wants to stop the scent from being dispersed, the user may adjust the cover **36** to the closed position **242**. In the closed position **242**, the apertures **38** are aligned with the solid portions **202** of the outer shell **190** of the absorbent basket **34**, which block the apertures **38** and, consequently, block or prevent fluid communication with the absorbent basket **34**. In this way, air flowing through the air register assembly **10** does not carry the scent or carries a minimal amount of the scent.

(49) The user may also limit the dispersion of the scent by adjusting the cover **36** to be partially opened, but not fully opened. In such examples, the cover **36** may be disposed between the opened position **240** and the closed position **242** to partially align the apertures **38** with the perforated portions **204**.

(50) Generally, the cover **36** is configured to rotate relative to the absorbent basket **34**. The internal projections **220** may be configured to slide along an outer surface of the rim **196**. Additionally or alternatively, the annular wall **212** of the cover **36** may be configured to rotate relative to the internal projections **220** and the closed end **216**. In such examples, the internal projections **220**

extend from the closed end **216** and remain stationary relative to the absorbent basket **38**. The annular wall **212** may be configured to rotate separately to open and close the apertures **38**.

(51) The rotation of the cover **36** about a first predefined degree of rotation adjusts the cover **36** between the opened and closed positions **240**, **242** to allow and prevent the dispersion of the scent, respectively. The rotation of the cover **36** about the first predefined degree of rotation may not rotate the engagement shaft **32**, allowing separate control of the scent and the damper adjustment system **112**. The cover **36** is also configured to rotate about a second predefined degree of rotation, which is larger or a greater degree of rotation than the first predefined degree of rotation. When the cover **36** rotates beyond the first predefined degree of rotation to the second predefined degree of rotation, the cover **36** is configured to rotate the absorbent basket **34**, which, consequently, rotates the engagement shaft **32** and the damper adjustment assembly **112** and results in the adjustment of the damper plates **136**, **138**.

(52) Referring to FIGS. **1-10**, the air register assembly **10** includes the knob assembly **24** for housing and dispersing the scented fluid. The air flowing through the air register assembly **10** from the HVAC is configured to flow through and/or adjacent to the absorbent basket **34** to carry the scent from the absorbing material **194** and into the interior compartment **50**. The dispersion of the scent into the interior compartment **50** may be adjusted through the opening and closing of the cover **36**. The type or strength of scent may also be adjusted by adding the scented fluid to the absorbing material **194** and through the interchangeable absorbent baskets **34**.

(53) Use of the present device may provide a variety of advantages. For example, the user may open and close the cover **36** to prevent and allow the dispersion of the scent from the air register assembly **10** and into the interior compartment **50**. Additionally, the absorbent basket **34** is reusable, allowing the user to add the scented fluid to the absorbent basket **34** to be used multiple times. Further, the scent is integrated into the air register assembly **10**, allowing the sent to be carried with the airflow from the HVAC system. Moreover, the integration in the air register assembly **10** provides a compact and more concealed solution for adding the scent to the vehicle **12**. Further, the user may snap engage the cover **36** over the absorbent basket **34**, as well as interchange the absorbent basket **34** to change the scented fluid. Moreover, the knob assembly **24**, while providing the scent, may also control the position of the barrel **18**, the position of the rotating vanes **22**, and the position of the damper plates **136**, **138**. Additional benefits or advantages may be realized and/or achieved.

(54) According to various examples, a vehicle air register assembly includes a housing defining an outlet. A barrel is disposed within the housing proximate to the outlet. The barrel is rotatably coupled to the housing. A frame is coupled to the barrel. Vanes are rotatably coupled to the frame proximate to the outlet. A knob assembly is coupled to the frame adjacent to the vanes. The knob assembly includes a support body defining an interior. Pivots extend from the support body to engage the frame. An engagement shaft extends through the interior of the support body. An absorbent basket is coupled to the engagement shaft. The absorbent basket is configured to absorb a fluid for providing a scent. A cover is selectively coupled over the absorbent basket. The cover defines apertures to provide fluid communication with the absorbent basket. Embodiments of the present disclosure may include one or a combination of the following features: the absorbent basket includes an outer shell having perforated portions and an absorbing material disposed within the outer shell; the outer shell is constructed of a plastic material and the absorbing material is constructed of at least one of fabric and foam; the cover is operable between an opened position providing fluid communication with the absorbent basket via the apertures and a closed position; the absorbent basket includes solid portions and perforated portions; the apertures are aligned with the perforated portions when the cover is in the opened position and aligned with the solid portion when the cover is in the closed position; the cover defines internal projections configured to engage a rim of the absorbent basket; the absorbent basket defines ribs; and the cover defines internal grooves configured to receive the ribs.

(55) According to various examples, an air register assembly for a vehicle includes a barrel and vanes operably coupled to the barrel. A knob assembly is operably coupled to the barrel and the vanes. The knob assembly includes a support body, at least one absorbent basket selectively coupled to the support body, and a cover selectively coupled over the at least one absorbent basket. The cover is configured to rotate between an opened position and a closed position. The opened position is configured to provide fluid communication with the at least one absorbent basket via apertures defined by the cover. Embodiments of the present disclosure may include one or a combination of the following features: the at least one absorbent basket includes an outer shell and an absorbing material within the outer shell; the outer shell includes perforated portions for providing fluid communication with the absorbing material; the apertures configured to align with the perforated portions when the cover is in the opened position; the at least one absorbent basket includes an outer shell with alternating solid portions and perforated portions; the apertures are configured to align with the solid portions when in the closed position and the perforated portions when in the opened position; the at least one absorbent basket includes a first absorbent basket and a second absorbent basket; the first and second absorbent baskets are interchangeable in the knob assembly; the at least one absorbent basket includes ribs; the cover includes grooves configured to receive the ribs to couple the cover to the at least one absorbent basket; each of the vanes and the support body are coupled to a link bar; rotation of the knob assembly drives rotation of the vanes via the link bar; a support vane extending normal to the vanes; and the support vane includes a first vane portion on a first side of the knob assembly and a second vane portion on a second side of the knob assembly.

(56) According to various examples, a knob assembly for a vehicle air register includes a support body defining an interior. An engagement shaft extends through the interior. An absorbent basket includes a base and an outer shell defining an interior. The absorbent basket is coupled to the engagement shaft. An absorbing material is disposed within the interior of the absorbent basket. A cover is selectively coupled over the absorbent basket. The cover defines apertures and is operable between an opened position and a closed position. Embodiments of the present disclosure may include one or a combination of the following features: the absorbent basket includes coupling projections extending from the base to snap engage the engagement shaft; the cover defines recessed regions extending from proximate an edge thereof to proximate a closed end; the apertures are defined in the recessed regions; the outer shell includes alternating solid portions and perforated portions; and the apertures are aligned with the perforated portions when the cover is in the opened position and aligned with the solid portions when the cover is in the closed position.

(57) For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.

(58) It is also important to note that the construction and arrangement of the elements of the invention as shown in the exemplary examples is illustrative only. Although only a few examples of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes, and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connectors or other

elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system might be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary examples without departing from the spirit of the present innovations.

(59) Modifications of the disclosure will occur to those skilled in the art and to those who make or use the disclosure. Therefore, it is understood that the embodiments shown in the drawings and described above are merely for illustrative purposes and not intended to limit the scope of the disclosure, which is defined by the following claims, as interpreted according to the principles of patent law, including the doctrine of equivalents.

(60) It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present disclosure, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A vehicle air register assembly, comprising: a housing defining an outlet; a barrel disposed within the housing proximate to the outlet, wherein the barrel is rotatably coupled to the housing; a frame coupled to the barrel; a damper adjustment assembly; vanes rotatably coupled to the frame proximate to the outlet; and a knob assembly coupled to the frame adjacent to the vanes, wherein the knob assembly is configured to rotate about a horizontal axis to rotate the vanes and configured to rotate the barrel about a vertical axis, wherein the knob assembly includes: a support body defining an interior, wherein pivots extend from the support body to engage the frame; an engagement shaft extending through the interior of the support body; an absorbent basket coupled to the engagement shaft, wherein the absorbent basket is configured to absorb a fluid for providing a scent; and a cover selectively coupled over the absorbent basket, wherein the cover defines apertures to provide fluid communication with the absorbent basket, wherein the cover is configured to rotate a first degree of rotation about a rotational axis to rotate between opened and closed positions to control dispersion of the scent and further to rotate a second degree of rotation to rotate the engagement shaft to adjust the damper adjustment assembly between opened and closed positions.
2. The vehicle air register assembly of claim 1, wherein the absorbent basket includes an outer shell having perforated portions and an absorbing material disposed within the outer shell.
3. The vehicle air register assembly of claim 2, wherein the outer shell is constructed of a plastic material and the absorbing material is constructed of at least one of fabric and foam.
4. The vehicle air register assembly of claim 1, wherein the cover defines a plurality of recessed regions that alternate between solid regions and regions defining the apertures.
5. The vehicle air register assembly of claim 4, wherein the absorbent basket includes solid portions and perforated portions, and wherein the apertures are aligned with the perforated portions when the cover is in the opened position and aligned with the solid portion when the cover is in the closed position.
6. The vehicle air register assembly of claim 1, wherein the cover defines internal projections configured to engage a rim of the absorbent basket.
7. The vehicle air register assembly of claim 1, wherein the absorbent basket defines ribs, and wherein the cover defines internal grooves configured to receive the ribs.
8. An air register assembly for a vehicle, comprising: a barrel; a damper adjustment assembly;

vanes operably coupled to the barrel; a knob assembly operably coupled to the barrel and the vanes, wherein the knob assembly includes: a support body; at least one absorbent basket selectively coupled to a downstream edge of the support body, wherein the at least one absorbent basket includes a base forming an annular flange and ribs arranged along the base; and a cover selectively coupled over the at least one absorbent basket downstream of the support body, wherein the cover includes grooves configured to receive the ribs to couple the cover to the at least one absorbent basket, wherein the cover is configured to rotate between an opened position and a closed position, the opened position configured to provide fluid communication with the at least one absorbent basket via apertures defined by the cover, wherein the cover is configured to rotate a first degree of rotation about a rotational axis to rotate between opened and closed positions to control dispersion of the scent and further to rotate a second degree of rotation to rotate an engagement shaft to adjust the damper adjustment assembly between opened and closed positions.

9. The air register assembly of claim 8, wherein the at least one absorbent basket includes an outer shell and an absorbing material within the outer shell.

10. The air register assembly of claim 9, wherein the outer shell includes perforated portions for providing fluid communication with the absorbing material, the apertures configured to align with the perforated portions when the cover is in the opened position.

11. The air register assembly of claim 8, wherein the at least one absorbent basket includes an outer shell with alternating solid portions and perforated portions, wherein the apertures are configured to align with the solid portions when in the closed position and the perforated portions when in the opened position.

12. The air register assembly of claim 8, wherein the at least one absorbent basket includes a first absorbent basket and a second absorbent basket, and wherein the first and second absorbent baskets are interchangeable in the knob assembly.

13. The air register assembly of claim 8, further comprising: a link bar, wherein each of the vanes and the support body are coupled to the link bar, and wherein rotation of the knob assembly drives rotation of the vanes via the link bar.

14. The air register assembly of claim 8, further comprising: a support vane extending normal to the vanes, wherein the support vane includes a first vane portion on a first side of the knob assembly and a second vane portion on a second side of the knob assembly.

15. A knob assembly for a vehicle air register, comprising: a support body defining a hollow interior between an upstream support and a downstream support; an engagement shaft including a bracket coupled to the upstream support, an engagement end extending through the downstream support, and a solid body extending therebetween; an absorbent basket including a base and an outer shell defining an interior, the absorbent basket coupled to the engagement shaft, wherein the engagement end is coupled to the absorbent basket at a downstream edge of the support body; an absorbing material disposed within the interior of the absorbent basket; and a cover selectively coupled over the absorbent basket, wherein the cover defines apertures and is operable between an opened position and a closed position, wherein the cover is configured to rotate a first degree of rotation about a rotational axis to rotate between opened and closed positions to control dispersion of the scent and further to rotate a second degree of rotation to rotate the engagement shaft to adjust a damper adjustment assembly between opened and closed positions.

16. The knob assembly of claim 15, wherein the absorbent basket includes coupling projections extending from the base to snap engage the engagement shaft.

17. The knob assembly of claim 15, wherein the cover defines recessed regions extending from proximate an edge thereof to proximate a closed end, and wherein the apertures are defined in the recessed regions.

18. The knob assembly of claim 15, wherein the outer shell includes alternating solid portions and perforated portions.

19. The knob assembly of claim 18, wherein the apertures are aligned with the perforated portions

when the cover is in the opened position and aligned with the solid portions when the cover is in the closed position.
